

Answer Key — Mathematics Grade 10

Sub-Strand 3.1: Trigonometry I • Lesson 4: Tangent and the Sine–Cosine–Tangent Relationship:
Solving the Mango-Tree Problem

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	C	B	B	A	D	A

USE AFTER: PREDICT PHASE — Predict: mango-tree scenario think-pair-share (which ratio is most useful?)

Q1. A student in Kisumu measures the sun's angle with the ground as 53° . The shadow of a vertical mango tree on level ground is 9.0 m long. Which equation connects the angle, the shadow and the tree's height h directly?

- A. $\tan 53^\circ = 9.0 / h$
- B. $\tan 53^\circ = h / 9.0$**
- C. $\cos 53^\circ = h / 9.0$
- D. $\sin 53^\circ = h / 9.0$

Correct answer: B. The tree height h is opposite the 53° angle and the shadow is adjacent to it, so the ratio Opposite/Adjacent, which is $\tan 53^\circ$, links them. Sin would need the sun-ray length (hypotenuse), which is not given.

USE AFTER: OBSERVE PHASE — Part A: algebraic derivation of $\tan \theta = \sin \theta / \cos \theta$

Q2. In a right-angled triangle, $\sin \theta = O/H$ and $\cos \theta = A/H$. What is $\sin \theta \div \cos \theta$, that is $(O/H) \div (A/H)$, after simplifying?

- A. $(O \times A) / H^2$
- B. O / H
- C. O / A**
- D. A / O

Correct answer: C. $(O/H) \div (A/H) = (O/H) \times (H/A)$. The H cancels, leaving O/A , which is $\tan \theta$. Multiplying by A/H instead of inverting gives the wrong answer $(O \times A)/H^2$.

USE AFTER: OBSERVE PHASE — Part B: numerical verification table for 30° , 40° , 60° and 70°

Q3. From mathematical tables, $\sin 40^\circ = 0.6428$ and $\cos 40^\circ = 0.7660$. What is $\sin 40^\circ \div \cos 40^\circ$, to 3 decimal places, and which table value does it match?

- A. 1.41, which matches $\tan 40^\circ$
- B. 0.839, which matches $\tan 40^\circ$**
- C. 1.19, which matches $\tan 50^\circ$
- D. 0.49, which matches $\tan 40^\circ$

Correct answer: B. $0.6428 \div 0.7660 \approx 0.839$, which equals $\tan 40^\circ$ in the tables. Dividing cos by sin gives 1.19, multiplying gives 0.49 and adding gives 1.41, and none of these are $\sin \div \cos$.

USE AFTER: EXPLAIN PHASE — Part A: sensemaking discussion, Q1 (is tan unnecessary?)

Q4. We have proved that $\tan \theta = \sin \theta / \cos \theta$, so tan can be found from sin and cos. Why is the tangent ratio still useful?

- A. It is only useful when we know the hypotenuse
- B. It gives a direct route when we know an angle and one of the opposite or adjacent sides but not the hypotenuse**
- C. It is only useful for angles of 53°
- D. It is not useful, because sin and cos can always replace it

Correct answer: B. Tan uses only the opposite and adjacent sides, so it is a convenient shortcut when the hypotenuse is unknown or cannot be measured, as with a tree and its shadow.

USE AFTER: EXPLAIN PHASE — Part A: sensemaking discussion, Q2 (opposite or adjacent?)

Q5. The sun's rays make a 53° angle with level ground. A vertical tree of height h casts a 9.0 m shadow on the ground, forming a right-angled triangle. The 53° angle is at the tip of the shadow. Which side of the triangle is the 9.0 m shadow, relative to the 53° angle?

- A. The adjacent side, because it lies next to the 53° angle and is not the hypotenuse**
- B. It cannot be named until the tree height is known
- C. The hypotenuse, because it is the length we measured
- D. The opposite side, because it lies on the ground

Correct answer: A. The shadow forms one arm of the 53° angle and is not the side facing the right angle, so it is adjacent. The tree height is the side opposite the 53° angle, and the sun ray is the hypotenuse.

USE AFTER: EXPLAIN PHASE — Part B: mango-tree calculation, $h = 9.0 \times \tan 53^\circ$

Q6. A mango tree casts a 9.0 m shadow when the sun's angle with the ground is 53° . Using $\tan 53^\circ \approx 1.3270$ from tables, what is the height of the tree, to 2 decimal places?

- A. 0.15 m
- B. 10.33 m
- C. 6.78 m
- D. 11.94 m**

Correct answer: D. $\tan 53^\circ = h / 9.0$, so $h = 9.0 \times \tan 53^\circ = 9.0 \times 1.3270 = 11.943 \approx 11.94$ m. Dividing 9.0 by 1.3270 (6.78 m) is the inverted-ratio error.

USE AFTER: MODEL BUILDING PHASE — Model: add 'Ratio Web' and the Model Statement to the cumulative Trigonometry Model

Q7. The same 53° sun angle at the same time of day gives the same tangent ratio for a small pole and for a tall tree. Why does $\tan \theta = \text{Opposite/Adjacent}$ work for triangles of any size?

- A. Because the ratio Opposite/Adjacent stays constant for a given angle, whatever the size of the triangle**
- B. Because the ratio gets bigger as the shadow gets longer
- C. Because the opposite side is always equal to the adjacent side
- D. Because $\tan \theta$ depends on the length of the hypotenuse

Correct answer: A. Triangles with the same angles are similar, so their side ratios are equal. For a given angle, Opposite/Adjacent is fixed and does not depend on the triangle's size.